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Palaeomagnetism and geochronology of mid-Neoproterozoic Yanbian dykes, South China: implications for a *c.* 820–800 Ma true polar wander event and the reconstruction of Rodinia

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Abstract: We present new palaeomagnetic data from two generations of mafic dykes in the Yanbian region of the western South China Block, dated by the zircon U–Pb method at 824 ± 6 and 806 ± 8 Ma, respectively. The primary origin for the characteristic remanent magnetizations is supported by a positive baked contact test, a dyke-tilt test and rock magnetic data. After tilt corrections, 10 dykes from the *c.* 824 Ma group gave a mean remanent direction of $D = 230.1^\circ$, $I = -72.6^\circ$ with $k = 16.3$ and $\alpha_{95} = 12.3^\circ$, corresponding to a palaeopole at 42.5 °N, 131.8 °E with $A_{95} = 19.0^\circ$. Three dykes from the *c.* 806 Ma group give a mean direction of $D = 284.5^\circ$, $I = 42.6^\circ$ with $k = 76.5$ and $\alpha_{95} = 14.2$, corresponding to a virtual geomagnetic pole (VGP) at 18.2 °N, 31.0 °E with $A_{95} = 20.4^\circ$. After correcting for a 5° vertical-axis rotation of the study region, the two pole positions are at 45.1 °N, 130.4 °E and 14.1 °N, 32.5 °E, respectively. The *c.* 825–720 Ma palaeopole position from South China, East Svalbard and neighbouring continents fall on great circles on two alternative Rodinia reconstructions, possibly reflecting oscillating inertial interchange true polar wander events (IITPWs).

Although the existence of a Neoproterozoic supercontinent, Rodinia, is now widely accepted, the exact configuration of the components of the supercontinent is still a subject of debate (e.g. Li *et al.* 2008; Evans 2009). A key issue in this debate is the position of the South China Block (SCB) in the supercontinent (Li *et al.* 1995*b*; Zhou *et al.* 2002; Yu *et al.* 2008; Munteanu *et al.* 2010), although it has commonly been accepted that the SCB was part of Rodinia (e.g. Evans *et al.* 2000; Li *et al.* 2005; Li *et al.* 2008; Yu *et al.* 2008; Bogdanova *et al.* 2009). Therefore, new high-quality palaeomagnetic poles from South China should have implications for the configuration and history of Rodinia.

The mobility of the supercontinent Rodinia between 850 and 750 Ma is another contentious issue owing to the lack of adequate high-quality palaeomagnetic poles. Li *et al.* (2004) first suggested a common, approximately 90° rotation of Rodinia between *c.* 800 and *c.* 750 Ma, as indicated by data from South China, Australia, the Congo Craton and India. They interpreted such a rapid rotation as a true polar wander (TPW) event: that is, the rotation of the entire Earth's mantle and crust relative to its core and rotation axis (Kirschvink *et al.* 1997; Evans 1998). Maloof *et al.* (2006) reported new results from East Svalbard that could be interpreted to support such a hypothesis, but suggested there were, perhaps, repeated TPW events during the 820–750 Ma interval comparable to the so-called inertia interchange true polar wander (IITPW) (Kirschvink *et al.* 1997; Evans 1998). However, the East Svalbard dataset lacks direct precise age constraints other than chemostratigraphic correlations (Maloof *et al.* 2006; Swanson-Hysell *et al.* 2015).

The key to testing both models for Rodinia configurations and the TPW/IITPW hypothesis is to acquire more reliable palaeomagnetic data from well-dated Neoproterozoic rocks in continents that were integral parts of the supercontinent Rodinia. In this paper, we report new geochronological and palaeomagnetic results from the western SCB that refine its mid-Neoproterozoic apparent polar wander path. By comparing the SCB data with those of other continents, we will examine the position of South China in Rodinia, and the precise timing and rate of the possible TPW event(s).

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Geological setting

The SCB consists of the Yangtze Block in the NW and the Cathaysia Block in the SE (Fig. 1); these were joined together by *c.* 900 Ma (Z. X. Li *et al.* 2002; Li *et al.* 2009). Our study area, the Yanbian region, is located close to the present western margin of the SCB. It is bounded to the north by the 'Longmen Shan Thrust System' (LSTS), to the east by the 'Xianshuihe–Xiaojiang Transcurrent Fault System' (XXFS) and to the SW by the 'Red River Fault System' (RRFS) (Fig. 1), all major structures active during the Cenozoic (Burchfiel *et al.* 1995; Wang *et al.* 1998).

Precambrian rocks in the Yanbian region consist of four series: (1) early Neoproterozoic Yanbian Group metavolcanic and metasedimentary rocks (Li *et al.* 2006); (2) Neoproterozoic plutons and mafic dykes that intrude the Yanbian Group; (3) the Nanhua system, as represented by the patchy occurrence of the Lieguliu Formation (a local name for the more widespread Nantuo Formation) tillite unconformably overlying the Yanbian Group; and (4) the Neoproterozoic Sinian strata overlying both the patchy Lieguliu Formation and the Yanbian Group (Fig. 2). The earliest Neoproterozoic Yanbian Group is believed to have been deposited in a back-arc basin that was later converted into a foreland basin during the final assembly of the SCB (X. H. Li *et al.* 2002; Greentree *et al.* 2006). The Yanbian Group is strongly deformed as a result of the *c.* 1000–900 Ma Sibao Orogeny, which resulted from the collision between the Yangtze and Cathaysia blocks (Z. X. Li *et al.* 2002). The folds are typically upright tight folds, with roughly east–west-trending and subhorizontal fold hinges (Carter 2001; Li *et al.* 2006), indicating no obvious block tilting. The Yanbian Group is also variably metamorphosed, at least partly related to the mid-Neoproterozoic granitoids and mafic–ultramafic intrusions, including the *c.* 857 Ma (Li *et al.* 2003) Guandaoshan dioritic pluton, the *c.* 825 Ma Gaojiacun and *c.* 821–813 Ma Lengshuiqing mafic–ultramafic bodies (Zhu *et al.* 2006, 2007), and the *c.* 820–813 Ma Tongde dioritic and gabbroic intrusions (Sinclair 2001).

Mafic dykes in the Yanbian region (YBDs) are exposed north of the Gaojiacun and Lengshuiqing complexes. They generally trend ENE (Fig. 2), mostly subvertical, with thicknesses ranging from less than 1 m to tens of metres. The dykes are well preserved and were interpreted to be related to a Neoproterozoic intracontinental rift regime over a mantle plume (W. G. Zhu *et al.* 2008) (Fig. 2). W. G. Zhu *et al.* (2008) divided these dykes into two different generations based on both their

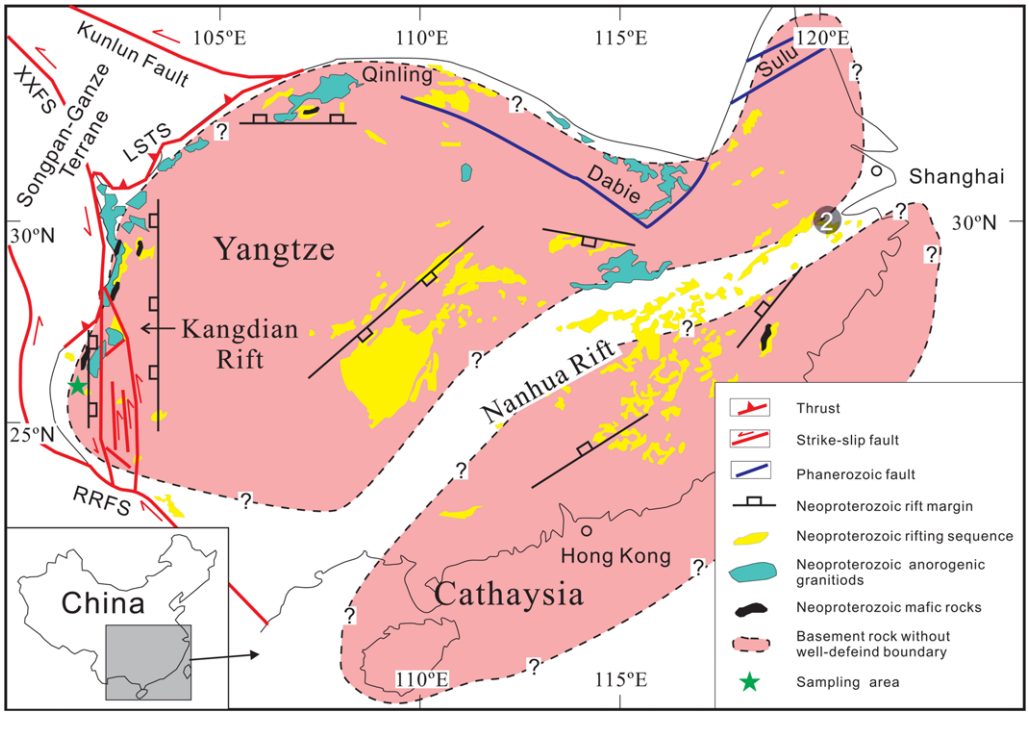


Fig. 1. Tectonic map of South China showing the studying area and the Neoproterozoic rift systems.

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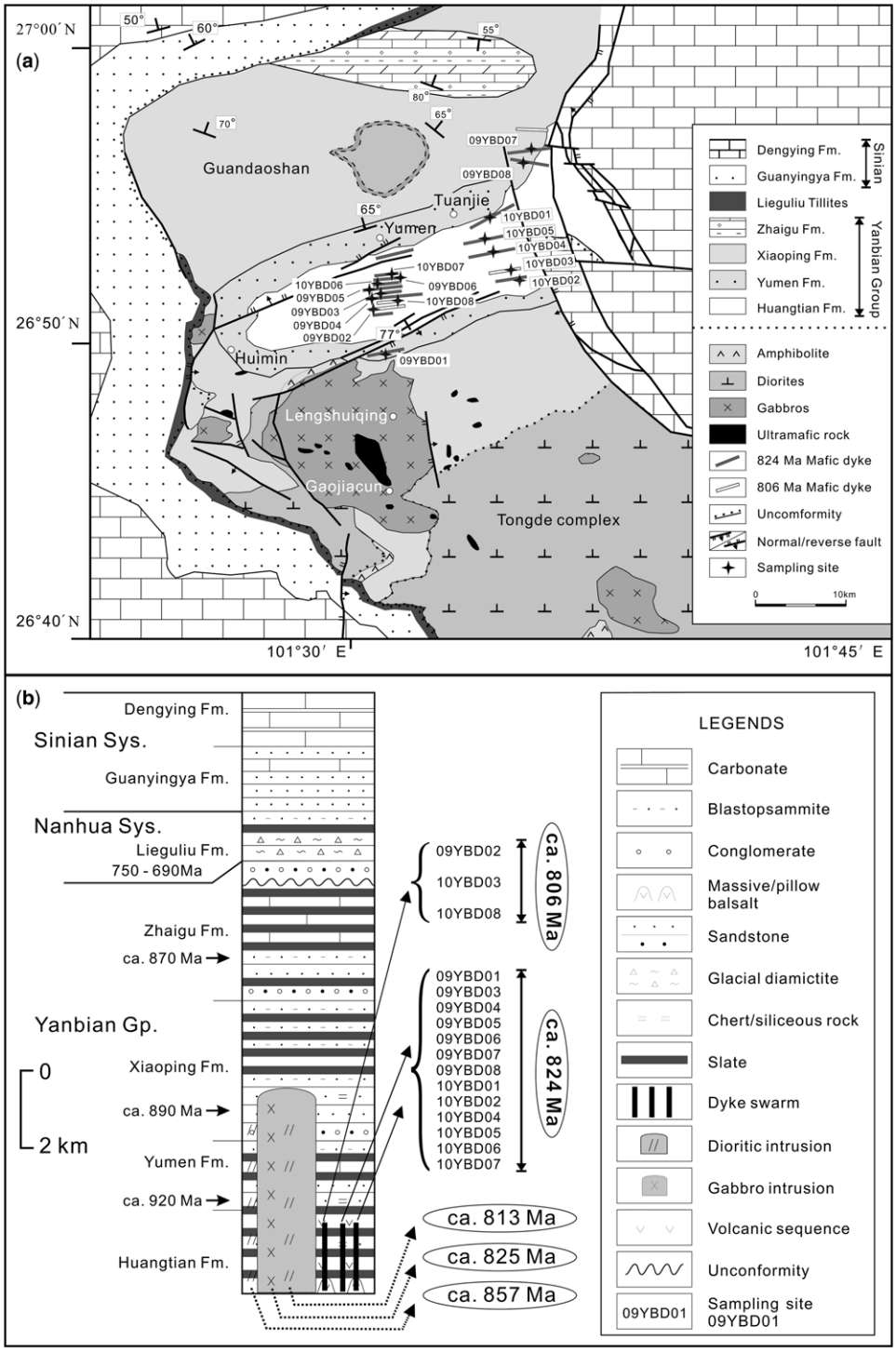


Fig. 2. (**a**) Simplified Precambrian geological map of the Yanbian area, SW China. Stars show the palaeomagnetic sampling sites (modified after the 1:200 000 geological map). (**b**) Proterozoic stratigraphy of the Yanbian region and contact relationships with the studied dykes. For references to isotopic ages, see the text.

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geochemical characteristics and preliminary age determinations. Their age determination was carried out using a Sensitive High-Resolution Ion Microprobe (SHRIMP) system located in Beijing, with two samples representing the two group of dykes giving ages of 792 ± 13 and 761 ± 14 Ma ages, respectively. However, these ages are no longer regarded as being reliable. The original data for sample HT-01 spread from 750 Ma to close to 900 Ma, and the authors interpreted older ages as being from zircon xenocrysts without giving supporting evidence. An inspection of the published cathodoluminescence (CL) images revealed no systematic differences between the two groups of zircon crystals. The original data for sample 04HS-01 mostly fall off the concordia line (see fig. 5b of W. G. Zhu *et al.* 2008). Both age samples were therefore re-analysed during this study through collaboration with W. Zhu.

Continental collision between India and Eurasia during the Cenozoic resulted in major faulting and block rotations in the region (Li *et al.* 1995*a*), featuring a series of north-trending, left-lateral faults as part of the XXFS (Fig. 1) (Wang *et al.* 1998).

Sampling and laboratory methods

A total of 122 orientated samples from 16 sites (09YBD01-08 and 10YBD01-08) were collected for palaeomagnetic analysis, mostly from mafic dykes but also a few from the adjacent country rocks (Fig. 2). Each site was from an individual dyke. The sampled dykes generally trend east–west and most dykes are vertical or subvertical. Dykes 09YBD02 (strike 96° and dip 50 °S) and 10YBD08 (strike 90° and dip 60 °S), both adjacent to faults at the SW edge of the swarm, have intermediate dips, the palaeomagnetic results of which may need tilt correction. No tilt reading for dykes 10YBD02 and 09YBD06 are available because of poor exposures. Some wall-rock samples were collected to perform a baked contact test. The wall rocks are part of the Huangtian Formation at the base of the Yanbian Group, consisting of metabasaltic pillow lavas interbedded with chert and schist. The sampling was carried using a portable diamond bit coring drill, and the samples were orientated with solar and magnetic compasses. Between seven and ten individually orientated palaeomagnetic samples with a standard 1 inch diameter were collected for each site (dyke), with at least one specimen from each sample progressively demagnetized using Magnetic-Measurements Ltd thermal demagnetizers and measured using a 2G three-axes cryogenic DC-SQUID (2G-755) magnetometer in a low-field coiled room at the Western Australian Palaeomagnetic and Rock Magnetic Facility in Perth. Heating was performed in 50°C increments to about 350°C, and then to a maximum temperature of 680°C with smaller temperature increments.

Remanence directions were estimated using the principal component analysis (PCA) method (Kirschvink 1980) in conjunction with inspection of orthogonal demagnetization plots (Zijderveld 1967) using the Super-IAPD software package (Torsvik *et al.* 2000). Site-mean directions were calculated by averaging the sample directions according to Fisher statistics (Fisher 1953), and the palaeopoles were analysed and plotted using GMAP software (Torsvik *et al.* 1990).

Anisotropy of low-field magnetic susceptibility (AMS) was measured using a Bartington MS2 meter to define the magnetic fabric of the rocks. Hysteresis and other rock magnetic properties were obtained using a Variable Field Translation Balance (VFTB) from Magnetic-Measurements Ltd. We used the ROCKMAG ANALYZER program (Leonhardt 2006) for plotting and interpreting magnetic properties such as magnetic carriers, coercivity, hysteresis and the Curie point.

The mount containing the two geochronological samples reported in was repolished, and the samples re-analysed for their zircon U–Pb ages using a Cameca IMS 1280 ion microprobe (SIMS) at the Institute of Geology and Geophysics, the Chinese Academy of Sciences (CAS) in Beijing. Zircons were documented with transmitted and reflected light micrographs, as well as CL images to reveal their external and internal structures. Detailed zircon U–Pb analytical procedures can be found in Li *et al.* (2009). The uncertainties in the ages for individual analyses are cited at the 1σ level, and the weighted mean ages are quoted at the 95% confidence level (*c.* 2σ).

Geochronology results

Zircon grains separated from dolerite dyke sample 09YBD02 (geochronological sample HT-01 of W. G. Zhu *et al.* 2008) exhibit simple prismatic igneous shapes and magmatic oscillatory zonings (Fig. 3b). Eighteen analyses of individual zircons were obtained (Table 1). Their U and Th contents vary from 20 to 426 ppm and from 7 to 295 ppm, respectively, with Th/U ratios of 0.06–1.38. Among the eighteen analyses, four gave significantly older ages (shown as grey ovals in Fig. 3a; Table 1), which are interpreted to be of xenolithic origin. The remaining well-clustered analyses gave concordant U–Pb ages within analytical errors (Fig. 3a), yielding a weighted mean $^{206}Pb/^{238}U$ age of 806 ± 8 Ma (MSWD = 1.6, 95% confidence interval). This age represents the crystallization age for dyke 09YBD02.

The geochronological sample from the centre of dyke 09YBD03 (geochronological sample 04HS-01

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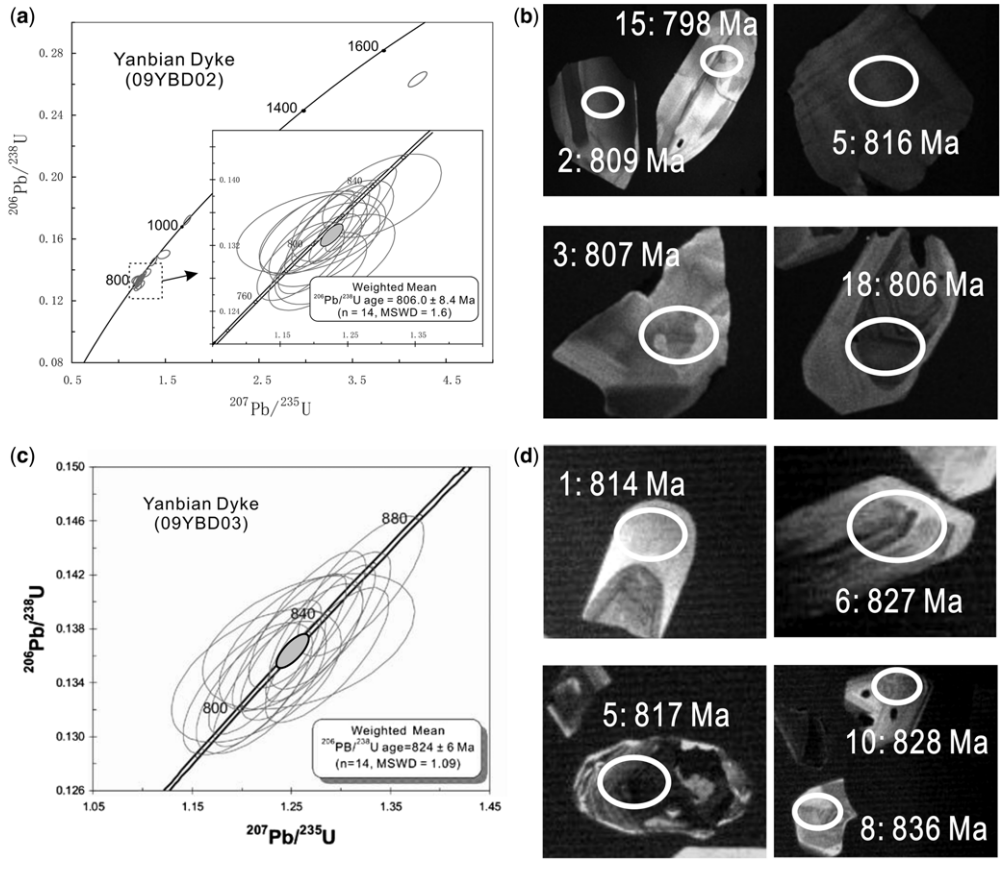


Fig. 3. Geochronology of Yanbian dykes 09YBD02 (geochronological sample HT-01 of W. G. Zhu *et al.* 2008) and 09YBD03 (geochronological sample 04HS-01 of W. G. Zhu *et al.* 2008): (**a**) U–Pb zircon concordia diagram and (**b**) CL images of zircon grains from 09YBD02; (**c**) U–Pb zircon concordia diagram and (**d**) CL images of zircon grains from 09YBD03. Analytical errors for individual analyses are at the 1σ level and the weighted mean ages are quoted at the 95% confidence level.

of W. G. Zhu *et al.* 2008) shows a gabbroic texture. Zircon grains are mostly euhedral to subhedral with internal growth zoning under CL (Fig. 3d). Fourteen analyses of individual zircons were obtained (Table 1). Measured U and Th concentrations vary from 46 to 294 ppm and from 21 to 251 ppm, respectively, with Th/U ratios of 0.43–1.04. All analyses gave concordant U–Pb ages within analytical errors (Fig. 3c), yielding a weighted mean $^{206}Pb/^{238}U$ age of 824 ± 6 Ma (MSWD = 1.1, 95% confidence interval). This age is interpreted as the crystallization age for dyke 09YBD03.

Palaeomagnetic results

Carriers of natural remanent magnetization

Of the 16 dykes studied, 13 exhibit characteristic remanent magnetizations of high laboratory unblocking temperatures. Representative samples for 13 sites were studied using the VFTB system with field intensities of between 1 and 60 mT and in normal atmosphere. The field intensity for each sample was chosen within this range in order to maximize the signal of the potential Hopkinson peak in the samples. If the external field is too high, it tends to saturate the magnetization of the sample and thus depress the potential Hopkinson peak.

Based on thermomagnetic behaviour, the samples are subdivided into three categories as described below.

Samples from three sites (09YBD02, 10YBD08 and 10YBD03) have Curie points of approximately 570–600°C, within 10–15°C of the Curie point of pure magnetite (Fig. 4a). These samples present magnetization peaks during the 500–600°C heating processes, but with enhanced magnetization intensity during the cooling cycles with no sign of such

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Table 1. *SIMS zircon U–Pb data for Yanbian dykes 09YBD02 and 09YBD03*

Spot	U	Th	Th/U	$f^{206}Pb$*	$^{207}Pb/^{206}Pb$	$\pm1\sigma$ (%)	$^{206}Pb/^{238}U$	$\pm1\sigma$ (%)	$^{207}Pb/^{235}U$	$\pm1\sigma$ (%)	$^{206}Pb/^{238}U$ age (Ma)	$\pm1\sigma$ (Ma)
Dyke 09YBD03												
1	46	21	0.46	0.19	0.0655	2.46	0.1345	1.50	1.2148	2.89	813.7	11.5
2	82	42	0.51	0.05	0.0662	1.61	0.1337	1.50	1.2196	2.20	808.7	11.4
3	242	251	1.04	0.00	0.0674	0.93	0.1377	1.51	1.2795	1.77	831.5	11.8
4	118	68	0.58	0.00	0.0671	1.32	0.1363	1.51	1.2604	2.01	823.6	11.7
5	170	77	0.45	0.08	0.0657	1.15	0.1351	1.51	1.2238	1.90	817.1	11.6
6	60	27	0.45	0.07	0.0662	1.85	0.1369	1.50	1.2498	2.38	827.3	11.7
7	127	54	0.43	0.05	0.0686	1.29	0.1361	1.50	1.2860	1.98	822.3	11.6
8	205	203	0.99	0.00	0.0675	1.23	0.1386	1.50	1.2891	1.94	836.8	11.8
9	86	45	0.52	1.79	0.0671	3.11	0.1371	1.50	1.2676	3.45	828.1	11.7
10	180	126	0.70	0.05	0.0661	1.10	0.1370	1.53	1.2480	1.89	827.5	11.9
11	205	173	0.84	0.09	0.0664	1.10	0.1343	1.52	1.2283	1.88	812.1	11.6
12	294	144	0.49	0.14	0.0678	1.29	0.1412	1.50	1.3202	1.98	851.2	12.0
13	193	86	0.45	0.73	0.0655	1.63	0.1336	1.50	1.2062	2.21	808.1	11.4
14	157	108	0.69	0.06	0.0662	1.17	0.1387	1.50	1.2661	1.90	837.2	11.8
Dyke 09YBD02												
2	174	197	1.13	0.01	0.0655	0.88	0.1338	1.50	1.2083	1.74	809.5	11.4
3	34	33	0.97	0.24	0.0652	2.27	0.1334	1.50	1.1991	2.72	807.2	11.4
4	214	295	1.38	0.04	0.0671	0.85	0.1347	1.50	1.2454	1.72	814.7	11.5
5	38	26	0.68	0.16	0.0668	1.98	0.1350	1.50	1.2426	2.48	816.3	11.5
6	81	57	0.70	0.09	0.0658	1.66	0.1301	1.50	1.1806	2.24	788.3	11.1
8	47	23	0.49	0.16	0.0685	2.39	0.1303	1.72	1.2313	2.95	789.7	12.8
10	78	55	0.71	0.08	0.0669	1.44	0.1310	1.51	1.2091	2.09	793.4	11.3
11	102	103	1.01	0.09	0.0661	1.20	0.1343	1.50	1.2249	1.92	812.5	11.5
12	26	22	0.85	0.05	0.0649	3.45	0.1336	1.52	1.1963	3.77	808.5	11.6
13	25	20	0.80	0.16	0.0683	2.32	0.1381	1.50	1.3007	2.76	833.7	11.8
14	42	26	0.62	0.00	0.0668	2.27	0.1321	1.58	1.2160	2.77	799.9	11.9
16	53	29	0.55	0.00	0.0647	1.85	0.1333	1.50	1.1893	2.39	806.7	11.4
17	362	217	0.60	0.07	0.0675	0.72	0.1364	1.53	1.2698	1.69	824.2	11.9
18	115	104	0.90	0.08	0.0679	1.95	0.1288	1.50	1.2057	2.46	781.2	11.0
1	62	43	0.69	0.04	0.0686	1.41	0.1453	1.50	1.3735	2.06	874.6	12.3
7	103	41	0.40	0.01	0.0731	1.10	0.1721	1.50	1.7338	1.86	1023.7	14.2
9	20	7	0.35	0.42	0.0717	3.27	0.1500	1.50	1.4830	3.60	900.8	12.6
15	426	26	0.06	2.37	0.1156	1.20	0.2632	1.52	4.1956	1.94	1506.1	20.4

*$f^{206}Pb$ is the percentage of common ^{206}Pb in total ^{206}Pb.

peaks. The peaks observed during the heating cycles could be interpreted as a Hopkinson peak. However, single-domain magnetite typically produces an inverse magnetic fabric (e.g. Potter & Stephenson 1988), which we do not observe here (Fig. 5). The observed peaks at 500–600°C are therefore more likely to indicate the generation of new magnetite, beginning at around 500°C, which soon became demagnetized when reaching the Curie point. Sample 10YBD08G plots in the pseudo-single domain field in the Day plot (Fig. 4d). Samples from one site (09YBD07) have a Curie temperature of approximately 530–550°C (Fig. 4b), suggesting titanomagnetite as the magnetic carrier. Samples from 12 sites (09YBD01, 09YBD03–09YBD06, 09YBD08, 10YBD01, 10YBD02 and 10YBD04–10YBD07) have Curie points of around 570–600°C, typical of magnetite. The flat heating curves before reaching the Curie point (Fig. 4c) indicate a high percentage of multi-domain or pseudo-single-domain magnetite as the remanence carrier. The Day plot (Fig. 4d) suggests that magnetite is dominantly in the pseudo-single-domain state.

Most samples show a dramatic increase in magnetization during cooling (dashed curves in Fig. 4a–c). This is probably due to the combined effects of demagnetization of single-domain magnetite during the heating runs that liberated the magnetite from its original magnetization, oxidation of titanomagnetite into magnetite and ilmenite, and/or the formation of new magnetite from minerals such as pyrite (Dunlop & Özdemir 1997).

Overall, both the rock magnetic results and the thermal demagnetization behaviour of the samples demonstrate that magnetite is the dominant carrier of remanent magnetizations in the studied

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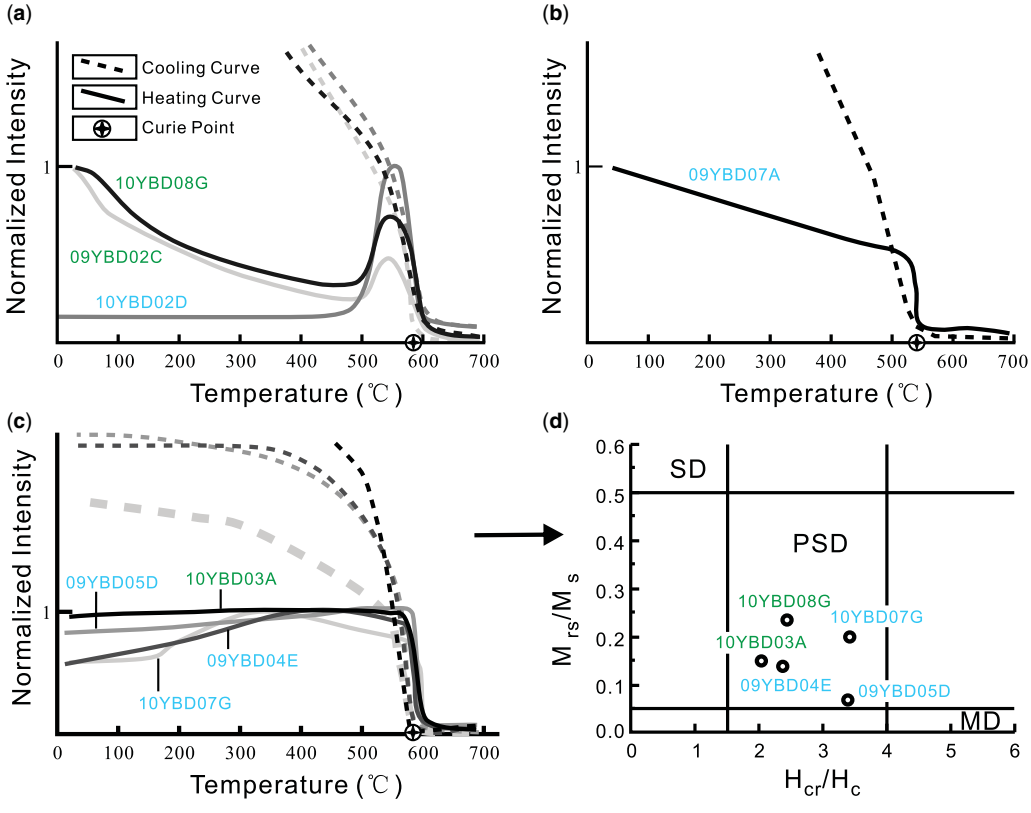


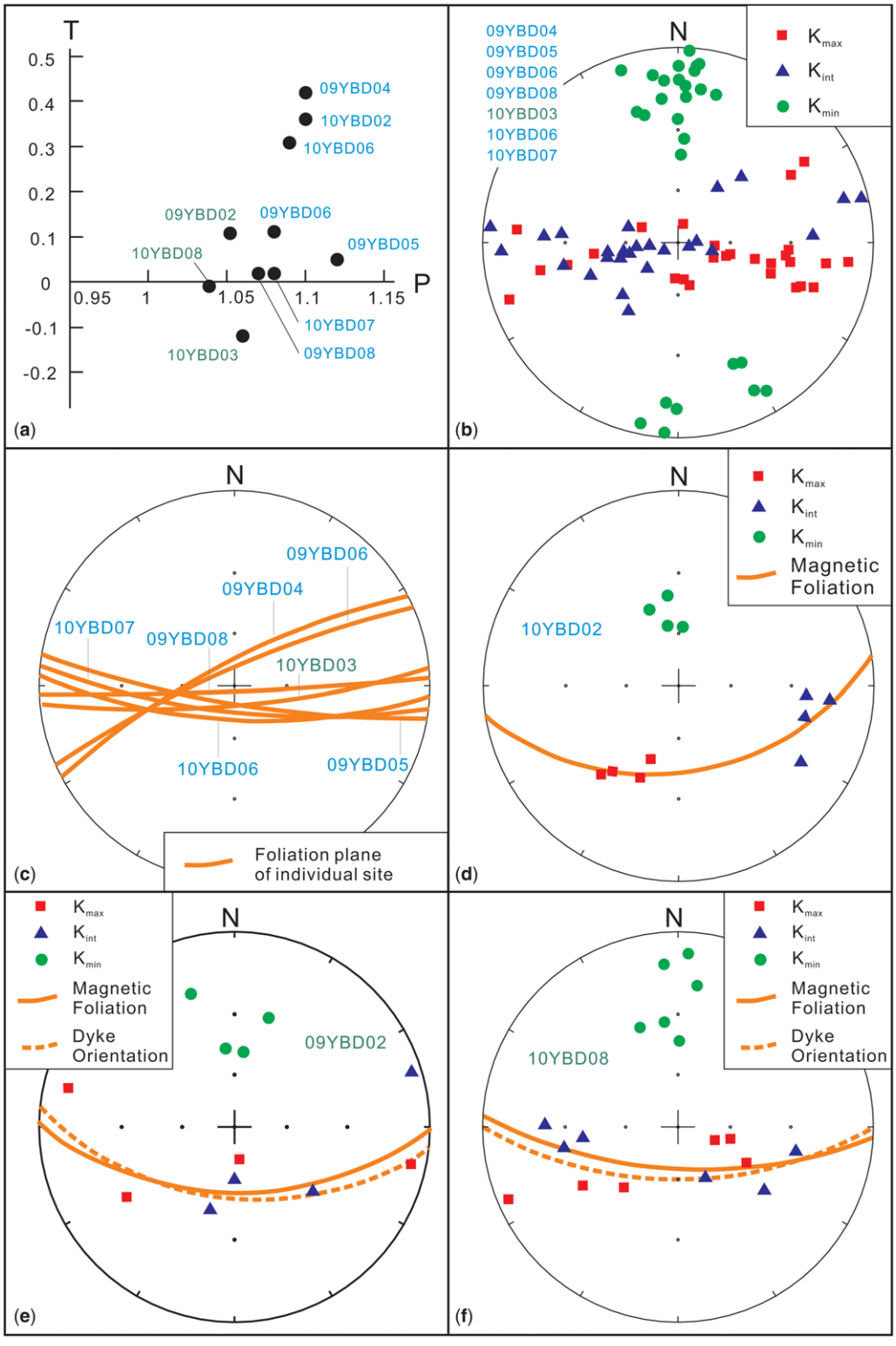
Fig. 4. (**a**) Thermomagnetic curves for samples with obvious peaks before reaching the Curie point. (**b**) Thermomagnetic curves for samples with titanomagnetite as the magnetic carrier. (**c**) Thermomagnetic curves for samples with flat thermomagnetic curves before the Curie point. The common approximately 15°C temperature gap around the Curie points between the heating and cooling runs in (a)–(c) was due to the samples delays in reaching the oven external temperature. (**d**) Day plot (Day *et al.* 1977) for sample type as in (c), as well as one sample from those in (a). Mrs, saturation remanent magnetization; Ms, saturation magnetization; Hc, coercive force; Hcr, coercivity of remanence; SD, single domain; PSD, pseudo-single domain; MD, multi domain. The numbers refer to the site of the sample. Plots are generated using RockMagAnalyzer 1.0 (Leonhardt 2006). Samples belonging to the 824 ± 6 Ma age group have their names shown in blue, and those belonging to the 806 ± 8 Ma age group have their names shown in green.

dyke samples, and the characteristic remanent magnetizations of high laboratory unblocking temperatures are most probably carried by pseudo-single-domain magnetite or titanomagnetite.

Anisotropy of magnetic susceptibility (AMS)

The degree of site-mean anisotropy ($P = K_{max}/K_{min}$) is typically ≤ 1.10 (Fig. 5a), except for three sites (09YBD07, 10YBD01 and 10YBD04), which showed no visible magnetic anisotropy and are excluded from further discussion in this section. Jelinek (1981) defined the T-factor to express the shape of the anisotropy ellipsoid. Oblate shapes correspond to $0 < T \leq 1$, while negative values, $-1 \leq T < 0$, correspond to prolate shapes. T values for most of the dykes (sites) are positive, indicating that the magnetic fabric of the samples is mainly oblate (Fig. 5a). The minimum principal axes (K_{min}) of seven sites (dykes) cluster in a north–south-directed shallow orientation, with the maximum and intermediate axes (K_{max} and K_{int}) defining an oblate magnetic fabric (Fig. 5b–c) parallel to the dyke orientations (Fig. 2), indicating that the dykes have a primary magnetic fabric (Jelinek 1981; Hrouda 1982). The foliation plane of dyke 10YBD02 strikes 81.1° and dips 42.5 °S (Fig. 5d) but the dyke orientation cannot be measured at the outcrop. We thus take the foliation plane as representing the dyke orientation. Dykes 09YBD02 and 10YBD08, both located at the southernmost part of the dyke swarm, are not subvertical as for most

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of the others, as is also shown by the dyke-parallel magnetic foliation (Fig. 5e–f).

K_{max} can be used to infer the magma flow direction (Knight & Walker 1988), but the K_{max} directions for most dykes are dispersed within the magnetic foliation without a clear concentration (Fig. 5b, e–f). Only site 10YBD02 has a well-clustered K_{max} direction (Fig. 5d), indicating a possible magmatic flow direction.

Most dykes that intrude the Yanbian Group dip subvertically (dip angles >75°), which is consistent with the AMS results (Fig. 5c). This implies that no tilt correction is necessary for remanence data from those dykes. Three dykes have shallower dips (strike 81° and dip 43 °S for site 10YBD02, strike 96° and dip 50 °S for site 09YBD02, and strike 90° and dip 60 °S for site 10YBD08: Fig. 5d–f). As all three dykes are located close to an ENE-trending thrust, their abnormal tilting angles could be related to post-intrusion tectonic tilting.

Palaeomagnetic results

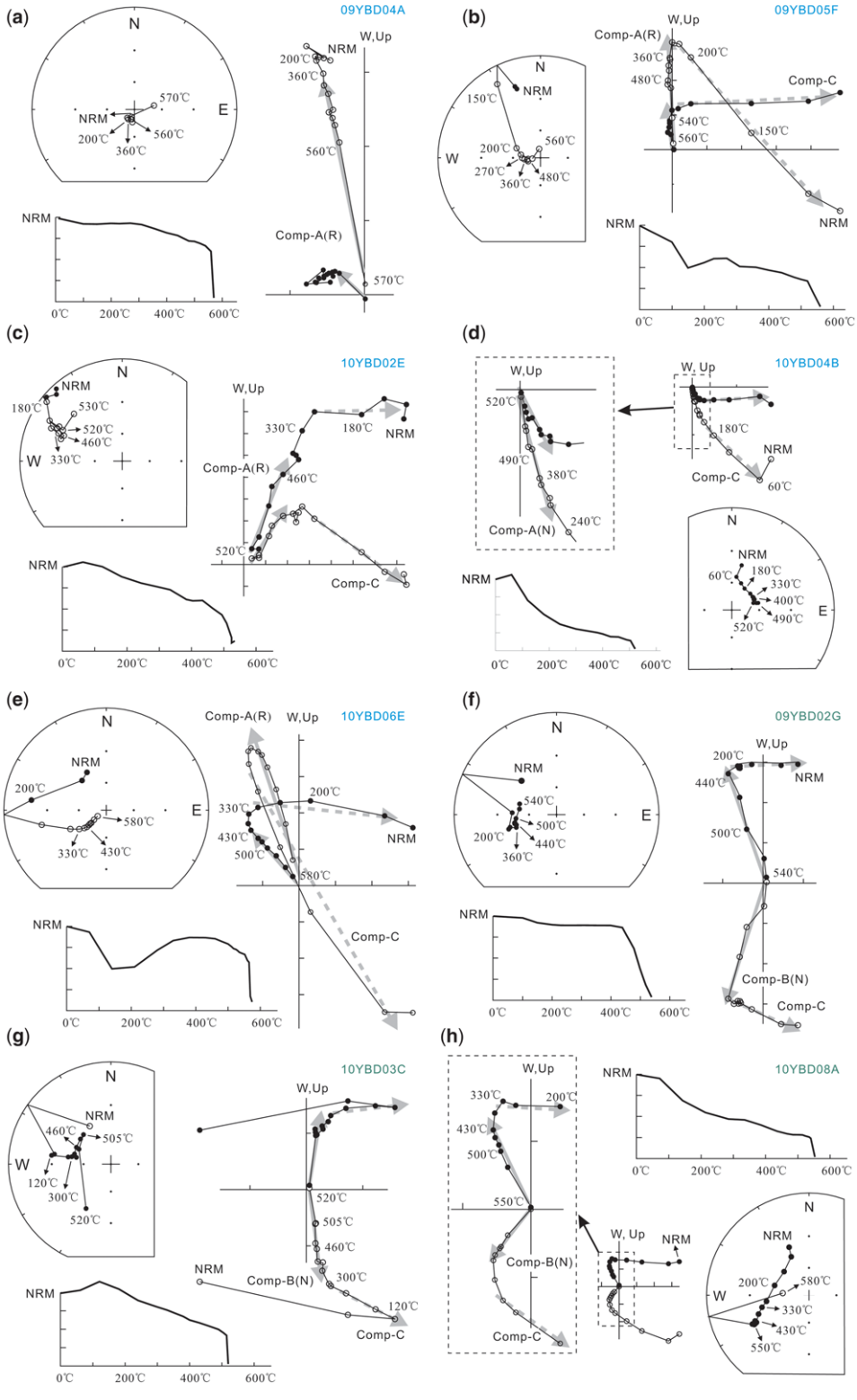
After thermal demagnetization, 13 of the 16 sites (each representing a single dyke) reveal high laboratory unblocking temperature magnetizations. We classify the high laboratory unblocking temperature remanence (the characteristic remanence (ChRM)) into two components, corresponding to the two age groups of the dykes. We note that the palaeomagnetic grouping of the Yanbian dykes is consistent with the geochemical classification of W. G. Zhu *et al.* (2008). Additional dykes not analysed by W. G. Zhu *et al.* (2008) are grouped according to their palaeomagnetic signature. Component 'A' is directed either steep/moderate-upwards to the SW (Fig. 6a–c, e) in present-day coordinates, or in the opposite direction (Fig. 6d). Component 'A' is present in 10 sites/dykes that belong to the 824 ± 6 Ma age group (original Group 2 of W. G. Zhu *et al.* 2008) (Figs 2 & 7a). Component 'B' is directed moderately downwards to the west (Fig. 6f–h) in present-day coordinates and is present in three sites/dykes that belong to the 806 ± 8 Ma age group (original Group 1 of W. G. Zhu *et al.* 2008) (Figs 2 & 7b). The directions and Fisher (1953) statistics of the characteristic remanence are given in Table 2. All 13 sites have α_{95} values below 16°. Most of these sites also show a low laboratory unblocking temperature component 'C', which is generally directed parallel to the present-day geomagnetic dipole field (Figs 6 & 7) and is interpreted as a recent viscous remanence with no geological significance.

Component 'A', preserved in dykes dated at 824 ± 6 Ma, has a laboratory unblocking temperature of 520–580°C. After tilt correction for the dyke at site 10YBD02, the mean direction for the 10 sites shifts from $D = 240.7°$, $I = 73.8°$, $k = 11.9$ and $\alpha_{95} = 14.7°$ to $D = 230.1°$, $I = 72.6°$, $k = 16.3$ and $\alpha_{95} = 12.3°$ (where D is declination, I is inclination, k is the Fisher precision parameter and α_{95} is the radius of the cone of confidence at a 95% probability level) (Fig. 7a, c). The increase in the value of k from 11.9 to 16.3, and a decrease in α_{95} from 14.7° to 12.3°, suggest that: (1) site 10YBD02, along with sites 09YBD02 and 10YBD08, was probably affected by the regional tectonic deformation that folded the Sinian and younger strata, and caused some faulting in the study region; and (2) the dykes acquired their remanence before the Phanerozoic deformation. The palaeopole position corresponding to the 10 site virtual geomagnetic poles (VGPs) (one from each dyke, with dyke 10YBD02 tilt-corrected), named pole-A, is at 42.5 °N, 131.8 °E with $A_{95} = 19.0°$ (where A_{95} is the radius of the 95% confidence circle for a palaeopole calculated using site VGPs).

Component 'B', isolated from dykes of the 806 ± 8 Ma group, has an unblocking temperature of 450–550°C. Two wall-rock samples of 10YBD03 (samples G and H) show a component 'B' remanence. Sample L, about 30 m from the dyke contact, shows a 'reverse' polarity in comparison to that of the dyke samples (Fig. 8c). A correction for tilt of two non-vertical dykes (site 09YBD02 with strike 96° and dip 50 °S, and site 10YBD08 with strike 90° and dip 60 °S) changes the mean direction of the component from $D = 261.8°$, $I = 40.0°$, $k = 17.66$ and $\alpha_{95} = 33.7°$ to $D = 284.5°$, $I = 42.6°$, $k = 76.5$ and $\alpha_{95} = 14.2°$ (Fig. 7b, d). The improved grouping (an increase in the value of k from 17.66 to 76.5 and a decrease in α_{95} from 33.7° to 14.2°) in component 'B' with tilt correction demonstrates that these dykes also acquired their remanence before the Phanerozoic

Fig. 5. Plots of data from anisotropy of magnetic susceptibility. (**a**) P–T diagram (Jelinek 1981) expressing the shape of the susceptibility ellipsoid: P denotes the degree of anisotropy; T expresses the oblate ($0 < T < 1$) and prolate ellipsoid ($-1 < T < 0$). (**b**) Examples of lower-hemisphere equal-area projection plots of downward directions of the maximum (K_{max}), intermediate (K_{int}) and minimum (K_{min}) susceptibility axes of samples from sites 09YBD04-06, 09YBD08, 10YBD03, 10YBD06 and 10YBD07. (**c**) Individual magnetic foliation of sites in (b). (**d**) Plot of susceptibility axes and magnetic foliation of site 10YBD02. (**e**) & (**f**) Plots of susceptibility axes, magnetic foliation and dyke orientation of sites 09YBD02 (e) and 10YBD08 (f), respectively. Names of samples belonging to the 824 ± 6 Ma age group are shown in blue, and those belonging to the 806 ± 8 Ma age group are shown in green.

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deformation. The mean of site VGPs for component 'B' gives the pole-B position at 18.2 °N, 31.0 °E with $A_{95} = 20.4°$.

Evidence for the primary origin of the characteristic remanence

The following evidence is interpreted to suggest a primary origin for the characteristic remanence (ChRM) found in the two groups of the Yanbian dykes. First, the fact that dykes dated at about 824 ± 6 and 806 ± 8 Ma yielded different remanent directions suggests that the region has not experienced regional remagnetization since *c.* 825 Ma. This is consistent with the unmetamorphosed nature of both the Neoproterozoic mafic dykes we examined and the Neoproterozoic strata exposed in the region. Second, rock magnetic results (Fig. 4) indicate that pseudo-single-domain magnetite or Ti-magnetite are the principal magnetic carriers for the ChRM, which requires reheating to near the Curie point of magnetite to remagnetize (Middleton & Schmidt 1982), and there is no geological evidence for such an event. Third, the improved grouping of the ChRM directions after tilt correction for both the 824 ± 6 and 806 ± 8 Ma dykes indicates that the ChRM components were acquired prior to the Phanerozoic deformation. Fourth, results from a baked contact test at site 10YBD03 (a *c.* 806 Ma dyke, about 6 m thick) are equivocal but can be interpreted as a positive test. As shown in Figure 8, dyke 10YBD03 is exposed between dykes 10YBD02 (about 2 m thick) and 10YBD04 (about 3 m thick), both are of *c.* 824 Ma ages. In this test, we treat both the wall rock (the Yanbian Group metasediments and metabasalts) and the two *c.* 824 Ma dykes as the country rocks to dyke 10YBD03. Among the eight baked wall-rock samples at site 10YBD03, five from the northern side of the dyke (samples G, H, I, J and L) show stable high laboratory unblocking temperature remanence, whereas samples A, B and K revealed no stable moderate- or high-temperature remanence probably as a result of the weathering and alteration of magnetic minerals. Component 'B' is isolated from sample G and H at the contact (Fig. 8c), which is consistent with that of the same-aged dyke samples. A ChRM component was also obtained from sample 10YBD03L, which was collected about 30 m to the north of the dyke, but with a direction close to the reverse polarity of component B from the dyke. Samples 10YBD03I and 10YBD03J have westerly and subhorizontal high laboratory unblocking temperature remanence vectors that are different from the component 'B' direction, but fall on the great circle between opposite polarities of the component 'B' direction (Fig. 8a, c). One way to interpret this result is that sample L was chemically remagnetized by the hydrothermal activity associated with the emplacement of the *c.* 806 Ma dykes and captured a polarity reversal after the dyke intrusion, and samples I and J (10 m away from the 5 m-wide dyke) were only partially remagnetized from their more stable remanence associated with the dyke intrusion. Alternatively, sample L retains a pre-dyke remanence, samples I and J were partially overprinted by the dyke intrusion, and samples G and H were completely remagnetized by dyke 10YBD03. In the latter case, this would constitute a positive baked contact test for the *c.* 806 Ma dykes. The interpretation is further strengthened if we consider the adjacent *c.* 824 Ma dykes (10YBD02 and 10YBD04) as part of the country rock for the *c.* 806 Ma dyke (10YBD03) and that the ChRM direction of the younger dyke is significantly different from that of the older ones (Fig. 8). We therefore interpret the magnetization age of remanence component 'A' as 824 ± 6 Ma and that of remanence component 'B' as 806 ± 8 Ma.

Discussions

Correction for a minor regional block rotation

Previous palaeomagnetic studies (e.g. Huang & Opdyke 1992; Li *et al.* 1995*a*), GPS observations (e.g. Zhang *et al.* 2004) and field structural measurements (Wang *et al.* 1998) suggest that, owing to the India–Asia collision in the Cenozoic, the western edge of the SCB experienced some vertical-axis block rotations relative to the central SCB. Pre-Cenozoic palaeomagnetic results from these regions (including the study region) thus need to be corrected for such rotations before being applied to the rest of the SCB.

Wang *et al.* (1998) demonstrated that the Xianshuihe–Xiaojiang Fault System (XXFS), which separates the study region from the rest of the SCB (Fig. 9a), is characterized by a series of left-lateral faults within the SCB with a total offset of about 60 km. Compared to the stable Eurasia Plate (which

Fig. 6. Representative progressive demagnetization behaviour of specimens from Yanbian dykes are shown in terms of orthogonal vector projections, stereographic projections and relative NRM intensity decay curve upon thermal demagnetization. Samples belonging to the 824 ± 6 Ma age group have their names shown in blue, and those belonging to the 806 ± 8 Ma age group have their names shown in green.

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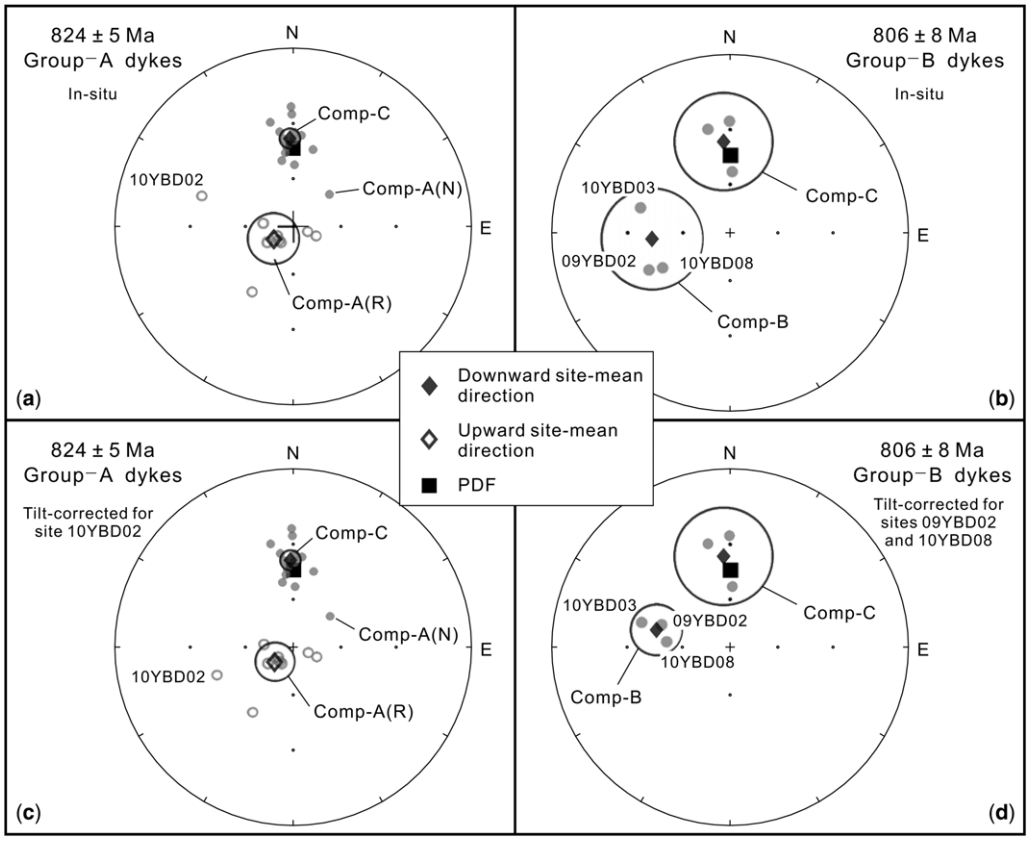


Fig. 7. (**a**) & (**b**) *In situ* normal (N) and reversed (R) site-mean directions of remanence components before tilt and foliation corrections. (**c**) & (**d**) Site-mean directions of remanence components after tilt and foliation corrections. Stereographic plots are in equal-angle projections, where the greyish open (closed) circles are projections on the upper (lower) hemisphere. Comp-A (-B and -C) represents site-mean direction of component 'A' (or 'B' or 'C'). PDF denotes the expected direction of the present dipole field. α_{95} is the radius of the cone of confidence at a 95% probability level.

the SCB is regarded as being part of since the late Mesozoic) reference poles (Besse & Courtillot 2002), palaeomagnetic results from 4.9 to 4.6 Ma sedimentary rocks at the base of the Yuanmou Basin (marked by a star in Fig. 9a) suggest a $5.3 \pm 3.0^{\circ}$ clockwise relative vertical-axis rotation for the region (Table 3) (Besse & Courtillot 2002; R. X. Zhu *et al.* 2008). Wang *et al.* (1998) and R. X. Zhu *et al.* (2008) demonstrated that the XXFS initiated in the early Pliocene, corresponding to the initial sedimentation of the Yuanmou Basin. The $5.3 \pm 3.0^{\circ}$ clockwise rotation should, therefore, reflect the total relative block rotation that the study region experienced due to the development of the XXFS. This estimated rotation magnitude is further supported by palaeomagnetic results of Palaeocene strata in the Yongren (marked as a solid diamond in Fig. 9a) and Dayao (solid pentagon in Fig. 9a) areas (Yoshioka *et al.* 2003), which indicate $10.3 \pm 8.4^{\circ}$ and $6.6 \pm 6.6^{\circ}$ clockwise rotations for the two areas, respectively, when compared with the 60 Ma palaeomagnetic pole for the Eurasia Plate (Table 3) (Besse & Courtillot 2002).

Using the total fault offset of about 60 km along the XXFS as the length of the small-circle arc defining the rotation, and the palaeomagnetically determined rotation angle (i.e. the declinational offset of *c.* 5.3°) as the angle of the rotation, we calculated the position of the Euler pole 'X' for this block rotation (Fig. 9b) at 25 °N, 96 °E (Fig. 9a).

Applying the 5° counterclockwise rotation around the Euler pole at 25 °N, 96 °E to the Yanbian region and the two palaeomagnetic poles from this study, the corrected pole positions are: pole-A at 45.1 °N, 130.4 °E with $A_{95} = 19^{\circ}$; and pole-B at 14.1 °N, 32.5 °E with $A_{95} = 20.4^{\circ}$ (Fig. 10; Table 2).

Table 2. *Palaeomagnetic data from Yanbian dykes (one site for each dyke)*

Site	Glat/Glon (°N/°E)	$n-n_o$	Direction				Cm	N/R/M	Palaeopole/VGP				
			Dec (°)	Inc (°)	k	α_{95} (°)			Plat (°N)	Plon (°E)	A_{95} (°)	dp (°)	dm (°)
09YBD04	26.856/101.517	7–8	220.5	−77.3	14.5	15.4	A	R	43.7	123.2		26.9	28.8
09YBD05	26.858/101.516	7–7	280	−71.2	63.8	7.6	A	R	16.7	136.9		11.6	13.3
10YBD06	26.858/101.517	5–6	228.8	−77.3	61.0	9.9	A	R	40.8	125.7		17.3	18.5
09BYD06	26.858/101.516	7–9	114.2	−73.8	22.7	13.0	A	R	35.1	67.5		21.1	23.4
10YBD07	26.860/101.515	7–7	109.7	−79.8	63.2	7.7	A	R	31.8	79.5		14.1	14.7
10YBD02	26.882/101.564	5–7	293.2	−33.6	51.4	10.8	A	R	−11.0	164.3		7.0	12.3
10YBD02#	26.882/101.564	5–7	252.3	−44.4	51.4	10.8	A	R	26.3	174.1		8.0	13.1
10YBD04	26.892/101.560	4–7	53.5	59.7	196.8	6.6	A	N	44.2	160		7.5	9.9
10YBD01	26.919/101.575	4–6	215.4	−43.2	131.5	8.0	A	R	58.2	186.6		6.2	9.9
09YBD08	26.934/101.584	6–6	239.6	−71.0	137.4	5.7	A	R	38.9	140.5		8.6	9.9
09YBD07	26.952/101.593	5–6	239.4	−78.4	65.8	9.5	A	R	36.2	125.5		16.9	17.9
Mean^{BTC}	26.886/101.546	10	240.7	−73.8	11.92	14.7	A	M	38.7	130.0	21.6		
Pole-A^{ATC}	26.886/101.546	10	230.1	−72.6	16.27	12.3	A	M	42.5	131.8	19.0		
Pole-A^{ARC}	26.886/101.546	10					A	M	45.1	130.4	19.0		
09YBD02	26.852/101.518	6–8	245.6	36.6	105.4	6.6	B	N	−10.9	41.1		4.5	7.7
09YBD02#	26.852/101.518	6–8	281.9	45.9	105.4	6.6	B	N	8.0	37.5		5.3	8.4
10YBD08	26.853/101.517	8–8	246.6	43.5	73.5	6.5	B	N	−7.3	44.9		5.0	8.1
10YBD08#	26.853/101.517	8–8	278.1	47.7	73.5	6.5	B	N	19.1	34.9		5.8	8.6
10YBD03	26.885/101.565	7–12	291.7	33.7	22.39	13.0	B	N	27.1	19.5		8.5	14.8
Mean	26.886/101.546	3	261.8	40.0	17.66	33.7	B	M	2.8	35.8	39.5		
Pole-B^{ATC}	26.886/101.546	3	284.5	42.6	76.52	14.2	B	N	18.2	31.0	20.4		
Pole-B^{ARC}	26.886/101.546	3					B	M	14.1	32.5	20.4		

Note: the site means are calculated from high-temperature components isolated in thermal demagnetizations. Negative value of Plat (N) represents an equal amount on the southern hemisphere. Glat/Glon, the geographical latitude and longitude of the sampling site; n, number of samples from a particular site that yielded the characteristic remanence; n_o, number of total sample number from that site; Dec, Inc, mean declination and inclination, respectively; α_{95}, radius of the cone of confidence at a 95% probability level; Cm, remanence component group; N/R/M, normal/reversed/mixed polarity of a remanence direction; Plat, Plon, latitude and longitude of the palaeomagnetic pole; A_{95}, radius of the 95% confidence circle for a palaeopole calculated using site VGPs; k, Fisher precision parameter; #, site mean after dyke tilt correction; $^{BTC}/^{ATC}$, pole calculated from site-mean directions after/before tilt corrections; ^{ARC}, after the 5° block rotation correction.

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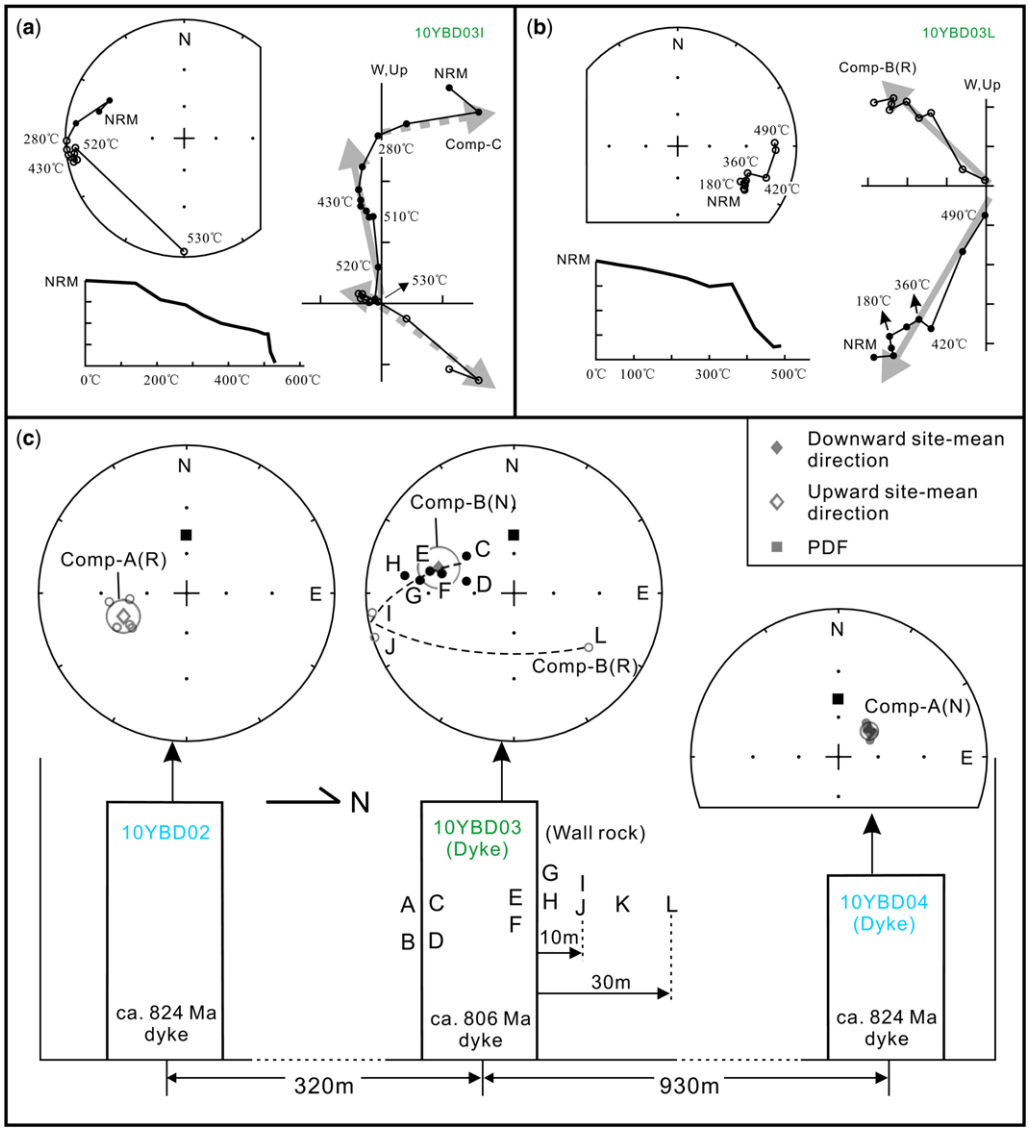


Fig. 8. (**a**) & (**b**) Progressive thermal demagnetization data of wall-rock samples 10YBD03I and 10YBD03L. (**c**) Diagram illustrating a field test at site 10YBD03 showing the positions of 11 samples. The thicknesses of 10YBD02, 10YBD03 and 10YBD04 are about 2, 6 and 3 m for each. Abbreviations are as in Figure 7. Samples belonging to the 824 ± 6 Ma age group have their names shown in blue, and those belonging to the 806 ± 8 Ma age group have their names shown in green.

Rodinia reconstruction and TPW

Numerous models have been proposed for the configuration of the Neoproterozoic supercontinent Rodinia, including the SWEAT (Hoffman 1991; Moores 1991), AUSWUS (Burrett & Berry 2000), AUSMEX (Wingate *et al.* 2002) and 'missing-link' (Li *et al.* 1995*b*, 2004, 2008) models. Although Li and co-authors (Li *et al.* 2008) favour the 'missing-link' connection for central Rodinia (see a revised model in Li & Evans 2011), due to the inadequate number of well-dated high-quality early-Neoproterozoic palaeopoles, the location of South China in Rodinia is still uncertain. Three of the most likely positions for the SCB are: (1) between western Laurentia and eastern Australia (Fig. 11);

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(2) adjacent to northern Australia (Evans 2000); and (3) adjacent to India and western Australia (Fig. 11b) (Jiang *et al.* 2003; Li & Evans 2011; Zhang *et al.* 2013).

We regard our new 824 ± 6 Ma palaeomagnetic pole-A as a high-quality pole, whereas the new *c.* 806 Ma pole-B can only be considered a virtual geomagnetic pole (VGP) as the small number of dykes that it is based on (three) is inadequate to average out the secular variation. While we have plotted both poles in Figure 11 for reference, in the following sections we will use pole-A only for the reconstruction of the SCB apparent polar wander path (APWP) and for palaeogeographical discussions. The new 824 ± 6 Ma palaeomagnetic pole-A is about 45° from the next younger palaeomagnetic pole for South china – the 802 ± 10 Ma Xiaofeng dykes pole (Li *et al.* 2004) (Fig. 11; Table 4). Pole-A places South China in the 50–80° palaeolatitude range. We use this pair of poles to re-evaluate the position of South China in Rodinia.

In Figure 11a, we reconstruct northern Rodinia as per the 'missing-link' model of Li *et al.* (1995*a*, *b*), taking into account the 40° rotation within Australia for pre-650 Ma poles (Li & Evans 2011) and other modifications as in Li *et al.* (2013) (see the Euler rotation parameters given in the caption of Fig. 11). In such a reconstruction, we align the paired 824 and 802 Ma poles from South China with the poorly dated, *c.* 810(?) and *c.* 800(?) Ma pole pair from East Svalbard. However, by doing so East Svalbard is put in a position away from Laurentia, but closer to Baltica (Fig. 11a). Such a position for East Svalbard is not in accordance with the traditional ideas of an East Svalbard–Laurentia connection (Harland & Gayer 1972) or the existence of a Precambrian East Greenland–East Svalbard platform (Gee & Teben'kov 2004; Halverson *et al.* 2004).

By placing India off NW Australia following Li *et al.* (2013), we are able to place the *c.* 814 Ma(?) Harohalli dykes poles (Radhakrishna & Mathew 1996; Pradhan *et al.* 2008) close to that of the new 824 ± 6 Ma Yanbian pole-A. In such a configuration, all of the *c.* 780–720 Ma poles from Australia, India and Laurentia cluster together, and a common 800–720 Ma APWP goes past the *c.* 790 Ma(?) pole for East Svalbard. However, there are two lower-quality Australian poles that fall on this general APWP, but show some discrepancies in ages. One is the ≤825 Ma Wooltana Volcanics pole (McWilliams & McElhinny 1980), which is not generally regarded as a high-quality pole because of the complexity of the demagnetization behaviour and the lack of field tests; this pole overlaps with the *c.* 800 Ma poles from South China and East Svalbard rather than with our new *c.* 824 Ma Yanbian dykes pole-A. The other is the poorly dated Browne Formation pole (Pisarevsky *et al.* 2007), which overlaps the high-quality 780–720 Ma poles rather than the 825–800 Ma poles. This possibly reflects the poor age constraint for this clastic formation.

This revised 'missing-link' model provides a tight fit for this part of Rodinia, and is consistent with the mantle plume and continental rifting

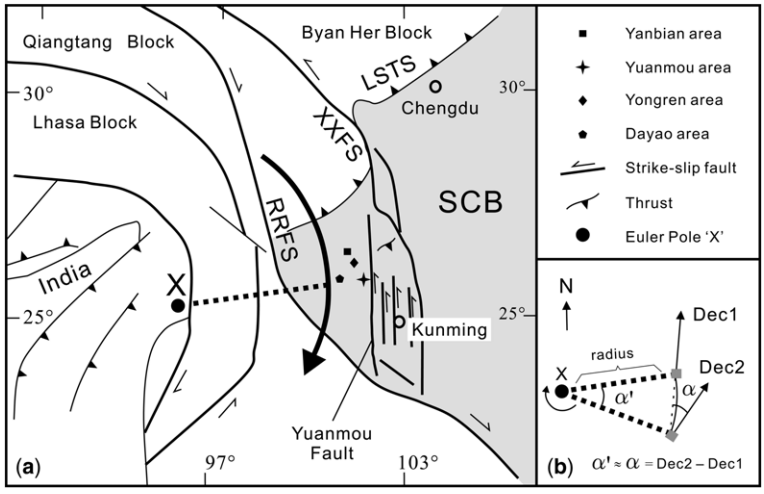


Fig. 9. (**a**) Sketch of the eastern Tibet Plateau geotectonics and the western SCB, and a generalized tectonic model relating displacement along the XXFS to rotation of crustal material. (**b**) Simplified model of rotation which indicates that the angle difference between declinations is equal to the rotational angle. Dec1/Dec2 denotes the declination before/after the rotation.

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Table 3. *Cenozoic palaeomagnetic directions and corresponding poles from the Yanbian–Yongren–Yuanmou–Dayao area*

Locality		Age	Observed directions			Pole position			Rotation	Reference
Name	Glat/Glon (°N/°E)		Dec (°)	Inc (°)	α_{95} (°)	Dec (°)	Inc (°)	A_{95} (°)	$R° \pm \Delta R°$	
Yuanmou	25.67/101.91	Pli	−2.8°	26.3°	1.6°	77.3°	297.3°	2.6°	5.3 ± 3.0	1
Yongren	26.02/101.63	Pal	20.2°	27.8°	8.3°	68°	218.5°	6.7°	10.3 ± 8.4*	2
Dayao	25.70/101.30	Pal	16.5°	31.3°	4.8°	72.3°	218.4°	4.5°	6.6 ± 6.6*	3

Note: tectonic rotation is reckoned with respect to Europe (Besse & Courtillot 2002). The amount of rotation (R) and its uncertainty (ΔR) are calculated using the method of Magill *et al.* (1981). We only chose the Palaeocene palaeomagnetic results from the Yongren area and recalculated a new Palaeocene pole.
*Rotation with respect to the declination expected from the pole of 60 Ma; Pli, Pliocene; Pal, Paleocene. Other abbreviations are as in Table 2.
Reference: 1, R. X. Zhu *et al.* (2008); 2, recalculated from Yoshioka *et al.* (2003); 3, Yoshioka *et al.* (2003).

records (e.g. Li *et al.* 2008). In Figure 11b, the SCB is placed adjacent to western Australia and southern India, and East Svalbard is placed adjacent to northern Laurentia. In such a configuration, the 825–720 Ma palaeopoles are in a similar level of agreement as in Figure 11a. However, this model leaves a gap between Australia and Laurentia in Rodinia.

In either case, all of the *c.* 825–720 Ma poles from South China, Australia, India, Laurentia

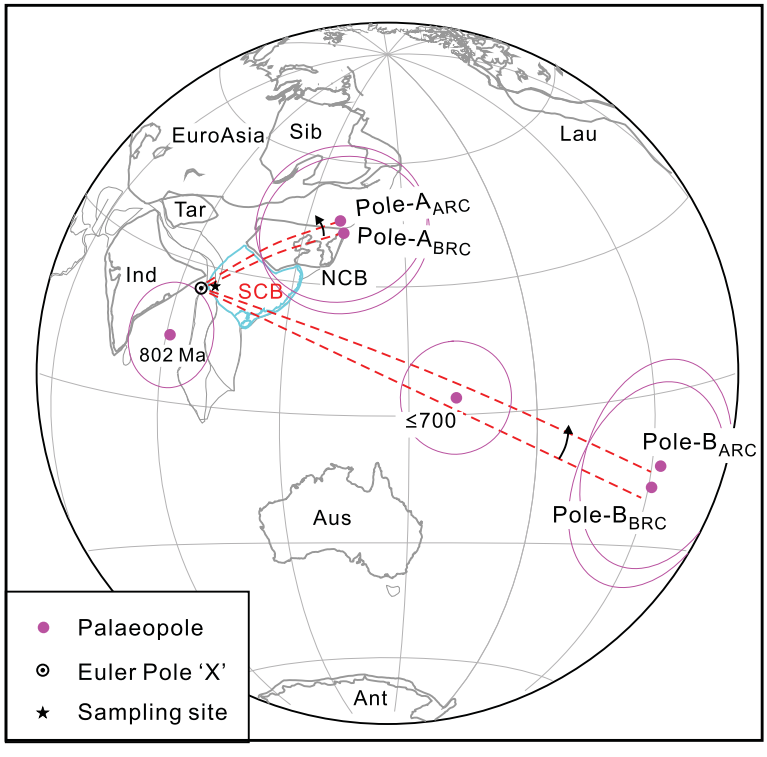


Fig. 10. Neoproterozoic palaeopoles in the Yanbian area are corrected counterclockwise around the Euler pole 'X' (25.0 °N, 96.0 °E) with a rotation angle of 5.3° in the present geography to account for Cenozoic clockwise deformation in the western margin of the SCB. Pole-A_{BRC} and Pole-A_{ARC} denote the position of pole-A before and after the rotational correction, respectively. Tar, Tarim Block; Sib, Siberia; NCB, North China Block; Ind, India Block; Aus, Australia; Ant, Antarctica.

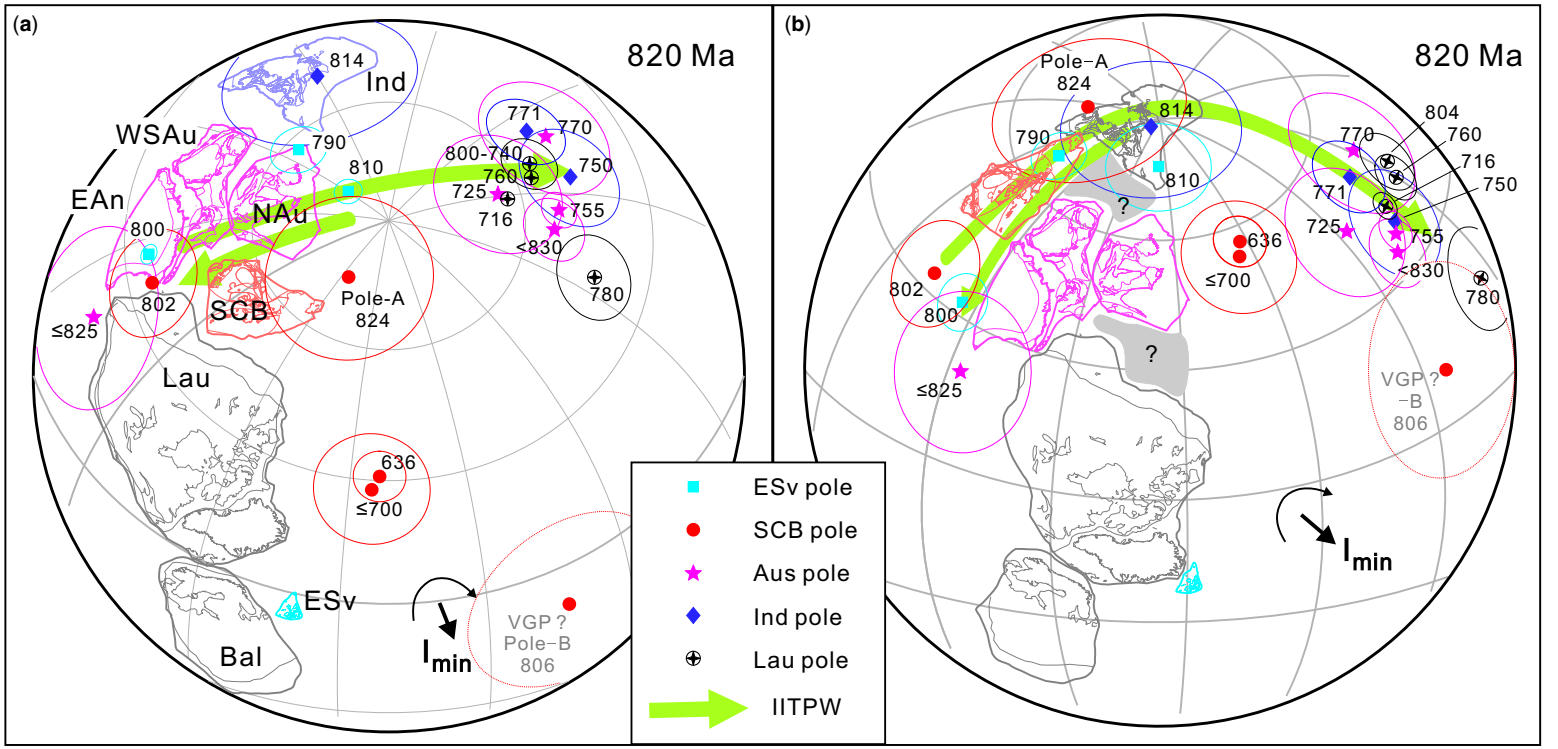


Fig. 11. Assessment of configurations of continents west and north of Laurentia at 800 Ma. Palaeopoles used are listed in Table 4. Owing to inferred intraplate rotation, southern Australia and East Antarctica was rotated counterclockwise relative to northern Australia (Li & Evans 2011) with the Euler pole of rotation of (−20.0, 135.0, 40.0). (**a**) SCB is assumed to be situated between Australia–East Antarctica and Laurentia. (**b**) Testing SCB's position adjacent to western Australia–East Antarctica. Euler poles used in (a) are: NAu (1.6, 207.1, −95.4), Bal (52.3, 232.7, 174.3), ESv (11.2, 305.2, 108.8), Ind (42.3, 60.3, 68.0), Lau (55.1, 209.6, −139.4), SCB (68.8, 76.1, 83.3). Euler poles used in (b) are: NAu (12.5, 24.8, 116.2), Bal (37.1, 253.5, 160.0), ESv (19.4, 313.2, 105.6), Ind (40.2, 19.3, 84.5), Lau (39.4, 229.9, −153.8), SCB (57.2, 106.3, 105.0). The opposite polarity of VGP Pole-B is displayed in the figure. Nau, north Australia; WSAu, western and southern Australia; Lau, Laurentia; Ean, East Antarctica; Bal, Baltica. Esv, East Svalbard; I_{min}, mimum inertia axis. Other abbreviations as in Figure 10.

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Table 4. *List of palaeomagnetic poles shown in Figure 11*

Rock unit	Pole name	Age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	Reference
Australia						
Walsh tillite	725	750–700	22.0	102.0	14.0	Li (2000)
Mundine Well dykes*	755	755 ± 3	45.0	135.0	4.0	Wingate & Giddings (2000)
Johnny's Creek Member	770	780–760	15.8	83.0	13.5	Swanson-Hysell *et al.* (2012)
Wooltana Formation*	≤825	825–800?	62.0	322.0	17.0	McWilliams & McElhinny (1980)
Browne Formation*	<830	830–800?	44.5	141.7	6.8	Pisarevsky *et al.* (2007)
East Svalbard						
Lower Svanbergfjellet	790	*c.* 790	−25.9	46.8	5.8	Maloof *et al.* (2006)
Upper Grusdievbreen	800	*c.* 800	1.1	72.6	6.1	Maloof *et al.* (2006)
Lower Grusdievbreen	810	*c.* 805	−19.6	24.9	11.0	Maloof *et al.* (2006)
India						
Mahe dykes, Seychelles	750	755–748	80.0	79.0	11.0	Torsvik *et al.* (2001)
Malani Igneous Suite	771	771 ± 9	67.8	72.5	8.0	Gregory *et al.* (2009)
Harohalli dykes	814?	814 ± 34	27.0	79.0	18.0	Radhakrishna & Mathew (1996)
Laurentia						
Franklin event	716	721–712	8.4	163.8	2.8	Denyszyn *et al.* (2009)
Uinta Mountain Group	760	780–742	0.8	161.3	4.7	Weil *et al.* (2006)
Gunbarrel dykes and sills	780	782–776	9.2	138.7	9.0	Harlan *et al.* (2008)
Galeros Formation	800–740	800–740	−2.1	163	6.0	Weil *et al.* (2004)
South China						
Nantuo Formation	636	636 ± 5	7.5	161.6	5.9	Zhang & Piper (1997); Zhang *et al.* (2008, 2013)
Liantuo Formation	≤700	≤700	4.0	161.0	13.0	Evans *et al.* (2000); Li *et al.* (2013); Lan *et al.* (2015)
Yanbian dyke – B	Pole-B 806	806 ± 8	14.1	32.5	20.4	This study
Yanbian dyke – A	Pole-A 824	824 ± 6	45.1	130.4	19.0	This study
Xiaofeng dyke	802	802 ± 10	14.0	91.0	11.0	Li *et al.* (2004)

Note: Owing to the intraplate rotation, Euler rotation (−20.0, 135.0, 40.0) is required for the Neoproterozoic poles from southern and western Australia (Li & Evans 2011), which is marked with * in the table.

and East Svalbard fall along a general circle of 90°–110° in length (Fig. 11). Li *et al.* (2004) first proposed a possible 90° TPW event between *c.* 800 and *c.* 750 Ma. Maloof *et al.* (2006) suggested a pair of TPW events at the time intervals of 810–800 and 800–790 Ma, with the later one thought to accord with the 800–750 Ma TPW event reported by Li *et al.* (2004). The *c.* 802 Ma Xiaofeng dykes pole (Li *et al.* 2004) and our new *c.* 824 Ma pole-A define an approximate 50° polar movement, a little too fast for plate tectonics (Meert *et al.* 1993). It is consistent with the 810–800 Ma TPW event speculated on the poorly dated East Svalbard results (Maloof *et al.* 2006). However, the *c.* 90° polar motion originally proposed for the *c.* 800–750 Ma interval (Li *et al.* 2004) more probably occurred between *c.* 800 and *c.* 780 Ma (Fig. 11). Our results therefore lend further support for two oscillating TPW events between *c.* 825 and *c.* 780 Ma during Rodinia time (Li *et al.* 2004; Maloof *et al.* 2006), probably IITPW events around a minumum inertia axis near the palaeo-equator (I_{min} in Fig. 11). Such IITPW events were possibly linked to the circum-Rodinia subduction and the consequent formation of antipodal mantle superplumes/superswells (Zhong *et al.* 2007; Li *et al.* 2008; Li & Zhong 2009), whereas the I_{min} axis near the palaeo-equator could reflect the relict mantle structure from the previous supercontinent cycle (Mitchell *et al.* 2012).

Conclusions

Palaeomagnetic and geochronological results from the Yanbian dykes in SW China reveal what are interpreted to be primary Neoproterozoic palaeomagnetic poles for the SCB: one 824 ± 6 Ma high-quality palaeopole; and a preliminary VGP of 806 ± 8 Ma. In combination with existing mid-Neoproterozoic palaeopoles from the SCB and other continents, we are able to demonstrate that the two major alternative positions for the SCB

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in Rodinia are still palaeomagnetically plausible: either between western Laurentia and eastern Australia in a 'missing-link' configuration (Fig. 11a); or adjacent to western Australia and southern India (Fig. 11b).

Approximately 825–720 Ma poles from South China, East Svalbard, Australia, India and Laurentia all distribute along a general circle 90°–110° in length (Fig. 11). In particular, the new *c.* 824 Ma pole and the existing *c.* 802 Ma pole from SCB indicate that an approximately 50° TPW event was likely to have occurred between 820 and 800 Ma, consistent with an early finding from East Svalbard (Maloof *et al.* 2006). This was followed by a possible approximate 90° TPW event in the opposite sense during *c.* 800–780 Ma (Fig. 11) (Li *et al.* 2004, 2013). Such oscillating TPW events can best be interpreted as inertia interchange true polar wander (IITPW) (Kirschvink *et al.* 1997; Evans 1998; Maloof *et al.* 2006).

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